

React.js: The Ultimate Beginner's Guide

1. What is React.js?

React is an open-source, front-end JavaScript library created by Meta for building user interfaces, specifically for single-page applications.

2. Core Concepts

- **Components:** Independent, reusable UI building blocks.
- **JSX:** Syntax extension allowing HTML-like structures in JavaScript.
- **Virtual DOM:** A lightweight copy of the real DOM for optimized updates.
- **Props & State:** Props pass data downwards; State manages internal component data.

3. A Simple Example

```
import React, { useState } from 'react';

function Counter() {
  const [count, setCount] = useState(0);

  return (
    <div>
      <p>You clicked {count} times</p>
      <button onClick={() => setCount(count + 1)}>
        Click me
      </button>
    </div>
  );
}
```

4. Why Choose React?

Feature	Benefit
Component-Based	Modular and maintainable code.
Declarative	Predictable and easy to debug.
Performance	Efficient rendering via Virtual DOM.